

Model performance — current state (2026-07-03)

temporal-exploit · numbers read **live** from artifacts/*.json at build time; the four pipeline-health items in section 4 are cited from the committed engineering log and marked **doc**. Backtest = 15-origin quarterly walk-forward, 1,310 test events, in-wild known-exploited target. **Honest caveat:** ~93% of merged in-wild labels are VulnCheck catalog-add dates (a publication-anchored known-exploited *proxy*, median lag ~175d), not exploitation onset — so this measures ranking against a catalog-membership label, not true in-the-wild timing.

1. Headline — structural XGB-AFT (EPSS-free) vs raw EPSS percentile

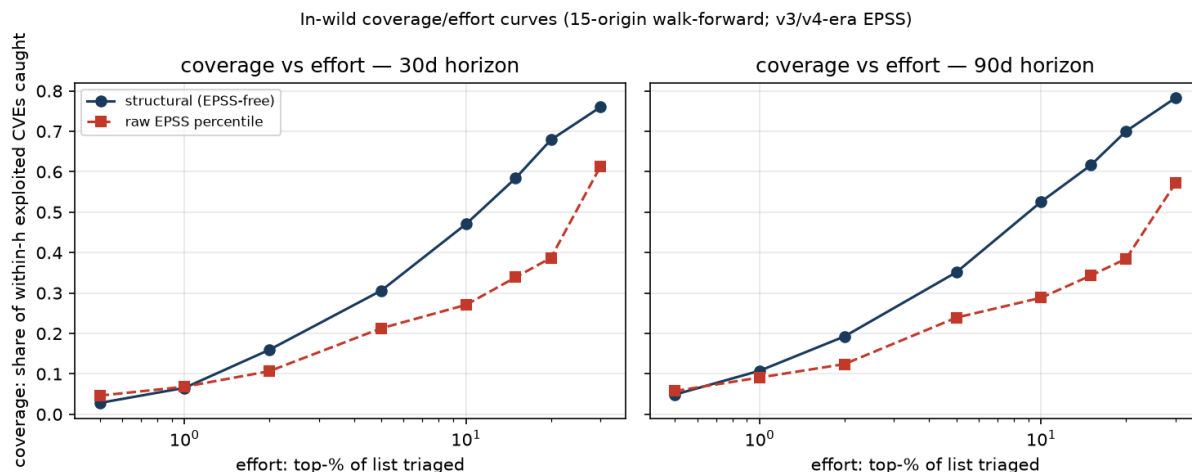
Metric	Structural	Raw EPSS	Δ structural – EPSS [95% CI]
Ranking AUC@30	0.795	0.695	+0.100 [+0.054, +0.146] (win 13/15)
Ranking AUC@90	0.816	0.683	+0.134 [+0.087, +0.180] (win 15/15)
Pooled PR-AUC@30 (n_pos 283)	0.0071	0.0040	+0.0031 [+0.0014, +0.0056] (99.8% of boots >0)
Pooled PR-AUC@90 (n_pos 483)	0.0134	0.0078	+0.0057 [+0.0020, +0.0088] (99.8% of boots >0)
Recall @ top-1% (30d)	0.065	0.068	EPSS marginally ahead at the very top of the list

Ranking is a clear win: structural beats raw EPSS by +0.100 [+0.054, +0.146] AUC@30 and +0.134 [+0.087, +0.180] AUC@90, both CIs well clear of zero (EPSS features are excluded by directive; the arm never sees an EPSS column). **PR-AUC verdict is now SUPERSEDED.** The prior standing note said “PR-AUC tied” — that rested on the per-origin paired delta, whose CI crosses 0 at only 15 origins (@30 +0.0050 [-0.0170, +0.0270]), i.e. underpowered, not a demonstrated tie. The NEW **paired pooled bootstrap** (pool all test rows across origins, resample; 283 positives @30 / 483 @90) shows structural wins PR-AUC **significantly** at both horizons: Δ AP CIs exclude 0 and 99.8% of bootstraps favour structural. **Keep the absolute scale in view:** the base rate is ~0.18%, so the AP values are tiny in absolute terms (0.0071 vs 0.0040 @30; 0.0134 vs 0.0078 @90). Read as a ratio the gap is material — structural carries ~1.79× the EPSS AP @30 and ~1.73× @90 — but this is a low-precision regime for both arms.

2. Coverage at fixed effort (fraction of true events caught in the top-k%)

Effort (30d)	Structural	EPSS	Δ struct – EPSS [95% CI]	Effort (90d)	Structural	EPSS	Δ struct – EPSS [95% CI]
0.5%	0.028	0.046	-0.018 [-0.051, +0.015] (win 2/15)	0.5%	0.049	0.058	-0.009 [-0.040, +0.022] (win 6/15)
1%	0.065	0.068	-0.003 [-0.026, +0.020] (win 5/15)	1%	0.108	0.091	+0.017 [-0.031, +0.065] (win 5/15)
2%	0.159	0.107	+0.053 [-0.008, +0.114] (win 8/15)	2%	0.193	0.124	+0.068 [-0.000, +0.137] (win 9/15)
5%	0.306	0.213	+0.093 [-0.013, +0.200] (win 10/15)	5%	0.352	0.239	+0.112 [+0.044, +0.181] (win 10/15)
10%	0.471	0.271	+0.200 [+0.104, +0.296] (win 12/15)	10%	0.526	0.288	+0.238 [+0.172, +0.303] (win 15/15)
20%	0.679	0.387	+0.292 [+0.153, +0.431] (win 13/15)	20%	0.700	0.385	+0.316 [+0.236, +0.396] (win 15/15)

How to read it (from the RE audit): at ≤1% effort the two arms are statistically indistinguishable — at 0.5% the point estimate actually favours EPSS at both horizons. The arms cross over between 1–2% effort. The structural advantage becomes significant (CI excludes 0) from **5% at 90d** and **10% at 30d**, and widens from there. The model's edge is a mid-list re-ranking benefit, not a top-of-list one — consistent with EPSS holding recall@top-1% in section 1.



3. This session's experiments — every one measured, verdict live from its artifact

Experiment	Numbers (backtest, live)	Artifact	Verdict
A3 · Seed variance (5 seeds)	AUC@30 sd = 0.000, AUC@90 sd = 0.000 (mean 0.795 / 0.816)	a3_seed_and_tune.json	Zero variance by construction. The config has no stochastic components (deterministic <i>hist</i> training) — the headline has no seed fragility.
A3 · Tuned config vs default	confirm-set Δ (winner–default): @30 +0.012 [-0.010, +0.033], @90 +0.016 [-0.005, +0.037]	a3_seed_and_tune.json	Default retained. 16 configs tuned on origins[:8]; winner's confirm-set deltas both cross 0 $\rightarrow$ no significant gain. Recorded as a candidate for when more origins accrue.
A2 · LambdaRank vs AFT	AUC@90 Δ (rank–AFT) -0.164 [-0.215, -0.113] (win 0/15)	a2_lambdarank_ab.json	Rejected. Pairwise rank objective loses decisively at every horizon; AFT stays the structural head.
XGB early stopping	wall 32.8s $\rightarrow$ 24.5s, but AUC@30 Δ -0.043 [-0.081, -0.006]	xgb_earlystop_ab.json	Rejected. ~25% faster, but tens of val events/origin make the stopping metric noisy and it costs accuracy. Ships opt-in only.
L1 · Vulnrichment SSVC labels	11 net-new CVEs vs KEV∪VulnCheck; 6 usable post-pub events; overlap lags existing by median +989d (0.18% earlier)	l1_vulnrichment_measure.json	Measured redundant — not wired. The last untested free historical source: SSVC assessments follow the catalogs, adding almost no CVEs and no timing gain.

4. Pipeline health (doc-cited from docs/improvement_log_2026-07-02.md)

Item	Measurement	Source
15-origin in-wild backtest	38.5 s $\rightarrow$ 35.15 s (~8.7% after the frame-hoist)	doc
Cached landmark-EPSS load	0.57 s vs ~4 min streamed (~450 $\times$) under the published-guard	doc
Peak RSS (81-col artifact build)	$\leq$ 1.46 GB against the 6 GB memory gate (~24%)	doc
Test suite	388 passed	doc
Adversarial RE audits	6 + 3 real breaks found and fixed across two rounds	doc

5. Standing statement — where the ceiling is, and what would move it

The in-wild ceiling is data-limited, and that is now fully verified. Every free historical label source has been measured: VulnCheck is integrated (it drove the 396 $\rightarrow$ 1,310-event lift); MSRC and Vulnrichment SSVC are both measured *redundant* (section 3); Shadowserver is ruled out (region/registration-locked); GreyNoise is verified prospective-only (rolling $\leq$ 30d, no historical backfill) and is accumulating. The model side is likewise exhausted — penalized Cox is the in-wild backbone, structural XGB-AFT is the EPSS-comparison headline, and cure / deep-survival / LambdaRank / resampling variants all measured negative.

What would actually move the needle next: (1) scheduled GreyNoise accumulation for onset-accurate prospective labels (the one path to true in-the-wild timing rather than a catalog proxy); (2) the deferred discrete-time head; (3) a NIST LEV baseline comparison (a citable afternoon of work); and (4) re-testing when a future EPSS model revision ships. EPSS scores come from the local `epss_history` file, which ends at the 2026-03-14 snapshot: v3/v4-era scores (v4 went live 2025-03-17 per first.org/epss/model). Any later FIRST model revision is out of scope of this history — comparisons are versus v3/v4-era EPSS. (An earlier 'v5 shipped 2026-06-15' claim did NOT verify against FIRST's site — RE audit 2026-07-03.)

Build: `scripts/build_performance_report_2026-07-03.py` · live sources: `inwild_epss_parity.json`, `inwild_remetric.json`, `a3_seed_and_tune.json`, `a2_lambdarank_ab.json`, `xgb_earlystop_ab.json`, `l1_vulnrichment_measure.json` + `docs/figures/fig_coverage_effort.png`; section 4 doc-cited from `docs/improvement_log_2026-07-02.md` · git branch master · generated 2026-07-03.